

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	10 July 2026
Team ID	
Project Name	Virtual Agent-Driven SLA Breach Awareness & Justification System
Maximum Marks	4 Marks

Technical Architecture:

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

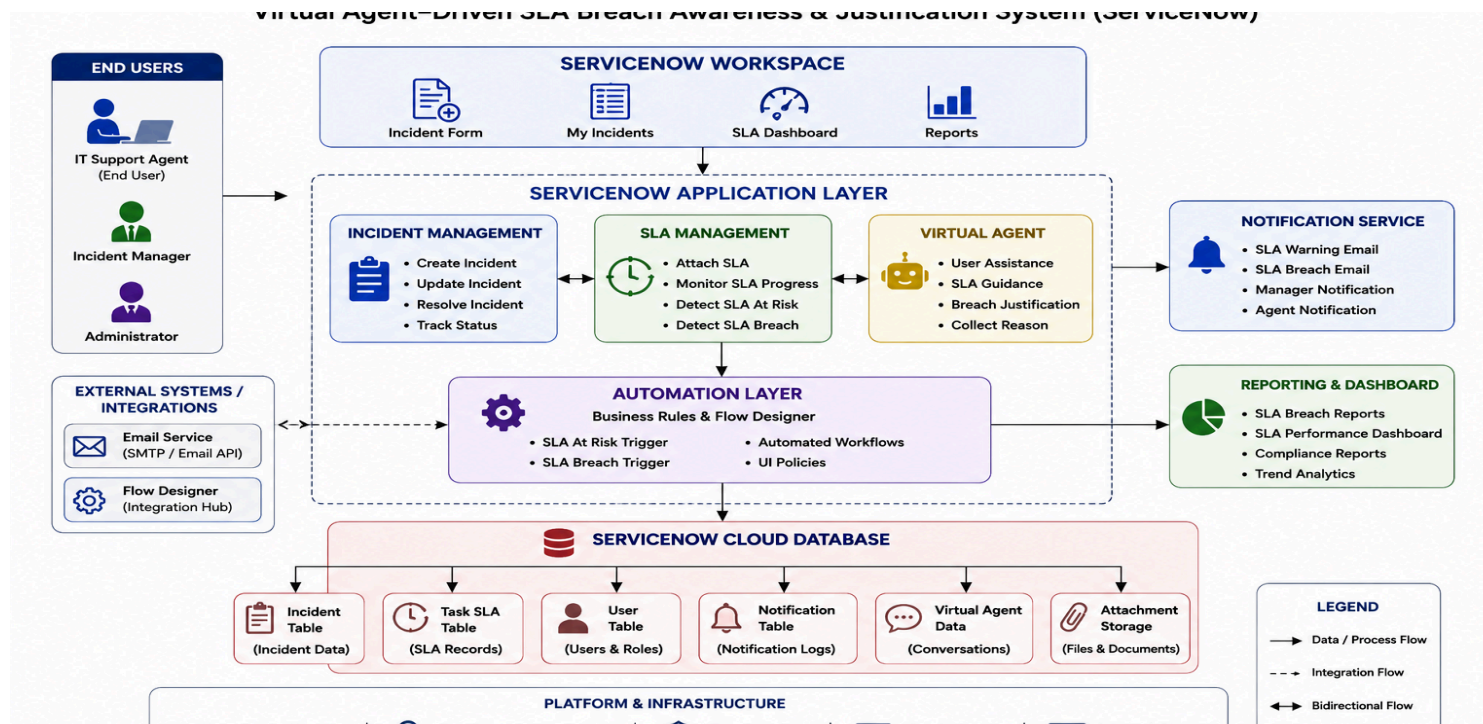


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Incident Workspace where Service Desk users create and manage incidents	ServiceNow Workspace UI
2.	Application Logic-1	Incident Management and SLA Monitoring	ServiceNow Incident Management
3.	Application Logic-2	SLA Attachment, SLA Tracking and Breach Detection	ServiceNow SLA Engine
4.	Application Logic-3	Virtual Agent Conversation and Justification Assistance	ServiceNow Virtual Agent
5.	Database	Stores Incidents, Users, Task SLA, Notifications	ServiceNow Tables
6.	Cloud Database	Cloud-hosted ServiceNow Platform Database	ServiceNow Cloud Database
7.	File Storage	Stores reports, attachments and logs	ServiceNow Attachment Storage
8.	External API-1	Email Notification Service	ServiceNow Notification Engine
9.	External API-2	Flow Designer Integration	ServiceNow Flow Designer
10.	Machine Learning Model	AI-based Virtual Agent conversation and response suggestions	ServiceNow Virtual Agent Intelligence
11.	Infrastructure (Server / Cloud)	Cloud hosted enterprise workflow platform	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No external open-source framework used. Project developed using ServiceNow platform components.	ServiceNow Platform
2.	Security Implementations	Role-Based Access Control (RBAC), User Authentication, ACLs, Secure Login	ServiceNow ACL, User Roles
3.	Scalable Architecture	Modular cloud architecture supporting multiple incidents and concurrent users	ServiceNow Cloud Architecture
4.	Availability	Highly available cloud infrastructure with continuous monitoring	ServiceNow Cloud Infrastructure
5.	Performance	Optimized SLA calculations, Flow Designer automation and indexed database tables	ServiceNow Platform Optimization

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>